

TOWHIDUL ISLAM

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EDUCATION

University of Twente

Enschede, The Netherlands

PgD in Geo-Information Science and Earth Observation

- GPA: 7.35/10 (Dutch Grade Class: 7)

Aug. 2021 – Apr 2024

Research: *Assessing Spatiotemporal Deformation Pattern of a Slow-Moving Landslide: Example From a Data-Sparse Region*

Erasmus University Rotterdam

Rotterdam, The Netherlands

MSc in Urban Management and Development - GPA: 6.60/10 (Dutch Grade Class: 7)

Sept. 2019 – Jan. 2021

Research: *Effects of Pluvial Urban Flood on Intracity Travel Behavior: The Case of Chittagong City, Bangladesh*

University of Chittagong

Chittagong, Bangladesh

MS in Geography and Environmental Studies (GPA: 3.92/4)

Jan. 2017 – Oct. 2018

Research: *Climatic Anomalies and Thunderstorm Intensity Over Bangladesh: A Spatiotemporal Analysis*

University of Chittagong

Chittagong, Bangladesh

BSc. in Geography and Environmental Studies (GPA: 3.64/4)

Feb. 2012 – May 2016

EXPERIENCE

SAR.Sense Geo-Intelligence Lab

Chittagong, Bangladesh

Founder & Co-ordinator

May 2025 – Present

- Established a research-driven lab with 3 specialized units (Disaster & Hazard Intelligence, Sustainable Agriculture & Forestry, Water & Environmental Intelligence), attracting **35+ team members and 5 external collaborators**.
- Coordinating 3 research projects as a lead researcher, delivering: 1. Explainable AI models for landslide susceptibility (SHAP/LIME), improving the interpretability of hazard maps. 2. A nationwide deformation atlas (2015–2025) using PSI/SBAS validated against **3+ GNSS stations**. 3. A compound-risk framework (InSAR + ABM) for Chittagong to simulate drainage and groundwater interventions, reducing modeled flood–subsidence risk by **up to 40%**.
- Architected and developed **BanGIS Pro (95% individual contribution)**, an AI-native geospatial & SAR analysis platform (FastAPI, React/TypeScript, Tauri) that lets users run complex multi-step remote-sensing workflows via natural language, without code — closing the gap between heavyweight GIS tools and code-heavy Python GIS stacks.
- Developed **Gridemy LMS**, a learning management system currently in active use by the Department of Geography and Environmental Studies, University of Chittagong, and under review for procurement by the Centre for Climate Change and Environmental Health (3CEH), Asian University for Women.
- Launched the *SAR.Sense Academy* with a vision to connect science with society through training and workshops on remote sensing, geospatial AI, and climate-risk modeling.
- Grow and coach a cross-functional team; run code reviews, documentation standards, ethical/data-governance practices, licensing/compliance, and partner/grant development with academia and agencies.

icddr,b

Dhaka, Bangladesh

GIS Consultant (National), Nutrition Research Division.

Jan. 2025 – May 2025

- Designed spatial sampling framework for the *One Nutrition Coverage Survey* with IFPRI & Johns Hopkins, **64** enumeration areas (EAs) across **4 districts** Dhaka, Sylhet, Khulna, and Rangpur.
- Built interactive survey maps integrated with Kobo, improving field data accuracy by **30%**.
- Trained **15 researchers** on GIS workflows, enhancing in-house spatial analysis capacity for the successful completion of the survey.

Red Cross Red Crescent Climate Centre

The Hague, Netherlands

Junior Researcher

Feb. 2021 – Jul. 2021

- Analyzed **30 years of meteorological data** (1990–2020) and conducted systematic review of literature and policy documents to establish Rajshahi's heat threshold (38°C).
- Led **remote sensing and GIS analysis** (Landsat, NDVI/NDWI, UHI mapping) to identify ward-level heat hotspots and drivers of urban heat risk.
- Supported **field validation and stakeholder interviews**, integrating local data with satellite analysis to ensure robust ground-truthing.
- Co-authored the study *"Identification of Heat Threshold and Heat Hotspots in Rajshahi, Bangladesh"*, commissioned under the ARRC programme and UK Met Office.
- Contributed to **policy recommendations adopted by Rajshahi City Corporation**, informing municipal climate resilience strategies and early warning protocols.

Halishahar School of Artillery

GIS Training Instructor

Chittagong, Bangladesh

Jan. 2017 – Aug. 2019

- Crafted and conducted a series of training programs on "Application of Geographic Information System in Military Intelligence and Counter Bombardment Cell Using ArcGIS" arranged for 20 army officers, achieving a 100% course completion rate.
- Utilized real-world case studies and active learning techniques, leading to a 40% improvement in the application of learned skills in practical situations over 1.5 years.
- Conducted tests before and after the training, observing a 60% rise in the trainees' ArcGIS competency from January 2017 to August 2019.

University of Chittagong

Research Assistant, Dept. of Geography and Environmental Studies.

Chittagong, Bangladesh

Apr 2017 – Aug. 2019

- Contributed to formulating a research proposal and a comprehensive project report.
- Collected, processed, and analyzed data on Bangladesh's lightning disaster trends, impacts, and management.
- Guided and monitored 6 undergraduate students related to this project, leading to a significant increase in project efficiency.
- Delivered progress reports and monthly updates for easy tracking and management of the project.

CURRENT PROJECT INVOLVEMENT

Modelling Deformation Susceptibility Using InSAR & XAI (Research Lead)

May 2025 – Present

- Develop explainable susceptibility models that fuse multi-temporal InSAR with geophysical, geological, and anthropogenic features; apply SHAP/LIME to rank risk drivers for planners.
- Operationalize an end-to-end pipeline (Sentinel-1 ingest → PSI/SBAS processing → feature engineering → model training/validation → maps/APIs) with QA and reproducibility.
- Deliver city-scale deformation-risk products (susceptibility maps, driver insights, confidence intervals) to inform infrastructure siting, monitoring, and early mitigation.

Nationwide Land Deformation Map of Bangladesh (Research Lead)

May 2025 – Present

- Assemble a national SAR archive (Sentinel-1, ALOS-2) and generate millimeter-level subsidence/uplift estimates via PSI/SBAS for 2015–2025, with GNSS/field validation.
- Publish a nationwide deformation atlas with interactive web layers and tiles for agencies (planning, water, transport) to prioritize resilience investments.
- Standardize metadata, uncertainty reporting, and change-detection dashboards to support policy, compliance, and climate adaptation programming.

InSAR–ABM Compound-Risk Framework for Chittagong

Jun. 2025 – Present

- Build an integrated framework to assess compound risk of land subsidence and pluvial flooding: multi-temporal InSAR (2015–2025) + rainfall/flood characterization + Agent-Based Model of household vulnerability and adaptation.
- Simulate intervention scenarios (e.g., drainage upgrades, groundwater management) and quantify risk-reduction efficacy; deliver compound-risk maps and a validated ABM for city planning.
- Produce actionable guidance and policy briefs for CDA/CCC/WASA and partners; package methods for replication in other high-risk cities.

SOFTWARE DEVELOPMENT & DIGITAL PRODUCTS

- BanGIS Pro — AI-Native Geospatial & SAR Analysis Platform** (*SAR.Sense Lab*) 2025 – Present
- *FastAPI · React/TypeScript · MapLibre GL · Tauri · NumPy/SciPy/GDAL · Groq LLaMA 3.3* — designed and built (**95% individual contribution**) a full-stack geospatial platform pairing a FastAPI backend and React/TypeScript/MapLibre GL client with a ReAct-style LLM agent that orchestrates multi-step GIS/SAR workflows from natural-language instructions, without code.
 - Implemented SAR/InSAR processing from first principles: radiometric calibration ($\beta^0/\sigma^0/\gamma^0$), Lee/Frost/Gamma-MAP speckle filtering, Range-Doppler terrain correction with Sentinel-1 orbit interpolation and DEM-based geocoding, and a full InSAR chain (interferogram generation, coherence estimation, phase unwrapping/displacement) feeding ~90 remote-sensing analysis tools.
 - Engineered a universal geospatial ingest pipeline (Shapefile, GeoTIFF, GeoPackage, NetCDF, HDF5, KML/GML) with automated CRS/geometry validation and a vector-tiling system rendering 250k+-vertex datasets in-browser; integrated Google Earth Engine for satellite data retrieval.
 - Validated with **438 automated backend tests** covering ingest, classification, georeferencing, and SAR/InSAR algorithms; positioned for global release as a free, natural-language alternative to costly commercial GIS software.
- Gridemy LMS — Learning Management System** (*SAR.Sense Geo-Intelligence Lab*) 2025 – Present
- Developed a learning management system currently in active use by the Department of Geography and Environmental Studies, University of Chittagong.
 - Under institutional review for procurement by the Centre for Climate Change and Environmental Health (3CEH), Asian University for Women, reflecting cross-institutional adoption of the platform.

SELECTED ACADEMIC PROJECTS

- Deformation Pattern Analysis of a Slow-moving Landslide** Sep. 2023 – Apr. 2024
- Utilized PSInSAR with multi-sensor SAR data (Envisat, Sentinel-1) to analyze spatio-temporal deformation patterns of a slow-moving landslide in a data-sparse region.
 - Integrated InSAR data with GRASS GIS geomorphological mapping to identify critical deformation zones and establish landform-deformation relationships.
 - Demonstrated the enhanced capability of multi-temporal Sentinel-1 data over Envisat in capturing finer-scale deformations, establishing a model for effective remote sensing in under-monitored regions.
- Soil-Plant Digital Twin Simulation using PySTEMMUS-SCOPE** Nov. 2023 – Feb. 2024
- Simulated land surface energy and soil moisture dynamics at Harvard Forest EMS Tower using PySTEMMUS-SCOPE within a Jupyter-based geospatial computing platform.
 - Analyzed seasonal variations in energy fluxes (Rnet, Qle, Qh, Qg), soil temperature, and soil moisture across vertical profiles and time series.
 - Interpreted model outputs to assess ecosystem responses to seasonal climate extremes; submitted processed visualizations and code to CRIB research infrastructure.
- Flood modelling (Physically-based)** Feb. 2023 – Mar. 2023
- Analyzed sequential levee breaches in same vs. different river systems (Maas and Waal) using LISEM flood modeling.
 - Simulated compound flood scenarios; evaluated flood arrival, depth, and velocity across spatial-temporal scales.
 - Delivered findings via report and presentation, highlighting implications for flood mitigation and evacuation planning.

TRAINING/SPECIAL COURSES/CONFERENCES

- Training Workshop: Turing MLxDL** Jun. 2020
Completed the Turing MLxDL Course offered by Turing Students Rotterdam. This course focused on advanced topics in Machine Learning (ML) and Deep Learning (DL).
- NASA ARSET Training: Using the UN Biodiversity Lab** Mar. - Apr. 2020
Completed the Turing MLxDL Course offered by Turing Students Rotterdam. This course focused on advanced topics in Machine Learning (ML) and Deep Learning (DL).
- MOOC- Studying Cities: Social Science Methods for Urban Research** Dec. 2019
Synthesized core social science research methodologies, focusing on mixed-methods application for urban problem-solving. This MOOC was offered by IHS Erasmus University on the Coursera platform.

NASA ARSET Training: SAR for Landcover Applications

Aug. - Sep. 2019

Completed the NASA's Applied Remote Sensing Training Program (ARSET) course on SAR for Landcover Applications, focusing on the use of Synthetic Aperture Radar (SAR) data for detailed land cover analysis.

NASA ARSET Training: RS for Monitoring Land Degradation and Sustainable Cities SDGs

Jul. 2019

Completed an Advanced Webinar by the NASA's Applied Remote Sensing Training Program (ARSET), focusing on utilizing remote sensing data for monitoring land degradation and contributing to Sustainable Cities SDGs.

NASA ARSET Training: Investigating Time Series of Satellite Imagery

Apr. 2019

Completed the NASA's Applied Remote Sensing Training Program (ARSET) course on Investigating Time Series of Satellite Imagery, which covered methods for analyzing temporal trends in remote sensing data.

Invited Talk: Spatiotemporal ... Bangladesh

Dec. 2018

Delivered an Invited Talk at the International Conference on Climate Change, Biodiversity and Sustainable Agriculture (ICCBSA-2018). The conference was jointly organized by Assam Agricultural University, Jorhat, India, and Prof. H.S. Srivastava Foundation For Science, Lucknow, India.

Training Course: Spatial Analysis Using 'R'

Dec. 2017

Participated in a training course on Spatial Analysis Using 'R' to develop advanced skills in spatial data processing and analysis. The course was organized by the Institute of Remote Sensing (IRS), Jahangirnagar University, Dhaka, Bangladesh.

Training Workshop on Google Earth Engine

Aug. 2017

Participated in a workshop on Google Earth Engine, a cloud-based platform for planetary-scale geospatial analysis. The training was jointly organized by the Institute of Remote Sensing (IRS-JU) and the International Centre for Integrated Mountain Development (ICIMOD), supported by NASA/SERVIR-HKH.

SOFTWARE & PROGRAMMING

ArcGIS Pro, QGIS, ERDAS IMAGINE, ENVI, SNAP, PostgreSQL, ILWIS, Python (GeoPandas, Numpy, Rasterio, Open3D, laspy, Shapely), R (Limited to statistical analysis), MATLAB (Basic), JavaScript (GEE), Tauri/React, Latex.

LANGUAGE PROFICIENCY

IELTS Academic (Aug. 2026) — Overall Band Score: **7.0** (CEFR C1)

Listening: 7.0 Reading: 7.5 Writing: 6.5 Speaking: 7.0

AWARDS

ITC Excellence Scholarship

Nov. 2021

Partially funded scholarship awarded by the Faculty of Geo-information Science and Earth Observation, University of Twente for studying Geo-information Science and Earth Observation at UT.

OKP Scholarship

Sep. 2019

Fully funded scholarship provided by Nuffic for MSc programme in Urban Management and Development at IHS, Erasmus University Rotterdam.

Fulbright Master's Fellowship

Jan. 2017

Fellowship awarded by the University of Chittagong for studying MS in Geography and Environmental Studies.

PUBLICATIONS

Journal Articles

Subedi, A., **Islam, T.**, Khan, R., Hassan, A., Hoogesteger, S. 2022. Identification of heat threshold and heat hotspot in Rajshahi, Bangladesh. Red Cross Red Crescent Climate Centre. Available at southasia.iclei.org

Sultana, N.N., Jabeed, A., Rahman, M.A., Hasan, S., **Islam, T.**, Podder, S., Mallick, S.S., Akter, H., Nath, T.K., Ullah, M.S., Paul, A. 2021. COVID-19 in Bangladeshi Daily Newspapers: A Thematic Analysis of Media Coverage. *Journal of Global Communication*, 14(2), pp.70–84.

Uddin, M. S., Mahmud, K., **Islam, T.**, & Rahman, M. A. (2026). Explainable machine learning assessment of hydroclimatic and land-surface controls on agricultural drought in North-Western Bangladesh. **Under review at** *Journal of Hydrology*.

Uddin, M. S., Mitra, B., Rahman, M. S., Mahmud, K., **Islam, T.**, Rahman, S. M., & Rahman, M. M. (2026). Integrating shared socioeconomic pathways and deep learning for future CO₂ emission forecasts in major European Union economies. **Under review** at *Environment, Development and Sustainability*.

Book Chapters

Uddin, M. S., Mahmud, K., Rahaman, M. A., Mitra, B., Rahman, S. M., **Islam, T.**, Rahman, M. M., & Rahman, M. S. (**Under Review**). Hands-on geomatics for marine conservation. In *Marine biodiversity dynamics in the Arabian Peninsula – A geomatics perspective of climatic and human impacts*. Springer.

REFERENCES

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